

YONAGO, Japan

WMO: 477440

Lat: 35.4347N

Lon: 133.3386E

Elev: 8

StdP: 101.23

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.2	-0.3	-6.3	2.2	3.4	-5.2	2.4	3.8	9.2	3.7	8.3	4.0	3.1	160	0.409

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.3	34.0	25.5	32.8	25.3	31.5	25.0	26.4	31.4	25.9	30.9	25.5	30.2	4.6	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.1	20.2	28.9	24.5	19.5	28.6	24.1	19.0	28.3	82.3	31.4	80.2	31.0	78.3	30.5	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.0	6.9	6.1	-3.2	36.3	1.4	1.0	-4.2	37.0	-5.0	37.6	-5.8	38.1	-6.8	38.8		
(5)			-4.0	27.2	1.3	0.5	-5.0	27.6	-5.8	27.9	-6.5	28.2	-7.5	28.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.6	4.6	5.1	8.1	13.2	18.3	22.0	26.6	27.7	23.2	17.7	12.3	7.2	(6)
(7)		DBStd	8.42	2.50	2.85	3.33	3.54	2.95	2.65	2.78	2.39	2.90	2.82	3.05	2.92	(7)
(8)		HDD10.0	502	168	139	78	10	0	0	0	0	0	0	12	95	(8)
(9)		HDD18.3	1883	425	369	316	158	37	2	0	0	1	46	183	346	(9)
(10)		CDD10.0	2530	1	3	21	106	258	359	513	549	396	238	79	7	(10)
(11)		CDD18.3	869	0	0	0	4	37	111	255	291	146	25	1	0	(11)
(12)		CDH23.3	7563	0	0	1	29	208	572	2488	3192	974	96	2	0	(12)
(13)		CDH26.7	2704	0	0	0	3	30	123	960	1326	249	13	0	0	(13)
(14)	Wind	WSAvg	2.8	2.9	2.8	3.0	3.0	2.9	2.5	2.6	2.7	2.6	2.5	2.6	2.8	(14)
(15)	Precipitation	PrecAvg	1860	150	136	124	118	121	193	237	146	227	135	130	126	(15)
(16)		PrecMax	2336	290	271	247	194	280	510	619	464	457	296	238	210	(16)
(17)		PrecMin	1237	52	36	54	44	46	56	7	8	29	41	48	42	(17)
(18)		PrecStd	276	49	44	39	37	51	101	157	97	115	66	44	43	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.6	18.0	21.7	26.3	29.3	31.7	35.0	36.1	33.5	28.5	23.2	17.6	(19)
(20)			MCWB	10.9	12.8	13.1	16.6	18.6	23.5	25.2	25.5	25.0	21.2	16.9	12.5	(20)
(21)		2%	DB	12.0	14.0	18.4	23.7	27.3	29.6	33.7	34.5	31.2	25.7	20.7	15.1	(21)
(22)			MCWB	7.9	9.8	11.6	15.4	17.6	22.3	25.3	25.6	24.5	20.1	15.6	10.8	(22)
(23)		5%	DB	10.4	12.1	16.2	21.6	25.7	28.0	32.5	33.2	29.3	23.9	19.2	13.6	(23)
(24)			MCWB	6.9	8.3	10.9	14.4	17.2	21.9	25.2	25.5	23.8	19.0	14.8	9.9	(24)
(25)		10%	DB	9.0	10.3	14.3	19.6	24.0	26.5	31.2	32.1	27.8	22.6	17.9	12.2	(25)
(26)			MCWB	5.8	7.1	9.9	13.6	16.8	21.1	24.9	25.3	23.0	17.9	13.9	8.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.3	13.6	14.7	19.3	21.7	25.0	26.7	27.2	26.2	22.8	18.5	14.1	(27)
(28)			MCDWB	14.1	17.3	19.1	22.9	25.3	29.4	31.8	32.4	30.4	26.8	21.1	16.1	(28)
(29)		2%	WB	8.8	10.6	12.9	16.8	20.0	24.0	26.1	26.6	25.2	21.4	16.9	11.7	(29)
(30)			MCDWB	11.1	13.0	16.3	21.9	23.8	27.8	31.3	31.7	29.4	24.6	19.6	14.3	(30)
(31)		5%	WB	7.3	8.9	11.6	15.5	18.9	23.0	25.7	26.2	24.5	20.0	15.6	10.4	(31)
(32)			MCDWB	9.8	11.4	15.0	19.8	23.3	26.5	30.7	31.1	28.4	22.7	18.4	13.1	(32)
(33)		10%	WB	6.1	7.3	10.4	14.4	18.0	22.1	25.2	25.7	23.8	18.7	14.3	9.1	(33)
(34)			MCDWB	8.7	9.8	14.0	18.1	22.4	25.3	30.1	30.6	27.2	21.5	17.3	11.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.3	7.1	8.6	9.5	9.1	6.9	6.7	7.3	7.1	8.3	8.0	6.8	(35)
(36)			MCDBR	8.9	11.0	12.4	12.8	12.0	9.2	8.7	8.8	8.7	9.7	10.1	9.5	(36)
(37)			MCWBR	6.1	7.5	7.6	6.7	5.6	3.9	3.1	3.2	3.7	5.0	6.0	6.3	(37)
(38)		5% WB	MCDBR	8.2	9.3	10.3	9.9	8.8	6.9	7.9	7.9	7.4	7.6	8.2	8.5	(38)
(39)			MCWBR	6.1	7.1	7.1	5.8	5.0	3.2	3.1	3.2	3.4	4.3	5.6	6.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.441	0.476	0.545	0.564	0.561	0.579	0.576	0.540	0.496	0.457	0.438	0.432	(40)	
(41)		taud	1.989	1.929	1.795	1.806	1.868	1.900	1.968	2.066	2.139	2.141	2.094	2.035	(41)	
(42)		Ebn at Noon	715	741	726	741	750	735	734	753	765	756	717	695	(42)	
(43)		Edh at Noon	138	164	205	212	202	196	182	162	144	131	124	125	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.72	2.45	3.59	4.74	5.23	4.88	4.83	5.03	3.89	3.17	2.17	1.56	(44)	
(45)		RadStd	0.14	0.25	0.32	0.39	0.50	0.38	0.72	0.56	0.34	0.28	0.18	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+70	N/A
(47)	Regional (23 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page